Regression Model

Anupong Banjongkan

Linear Regression

- Linear Regression is a simple and powerful model for predicting a numeric response
- From a set of one or more independent variables
- It is effective, easy to interpret, and easy to extend

Linear Regression

- Linear regression is a supervised algorithm
- It learns to model a dependent variable, y, as a function of some independent variables (aka "features"), x_i, by finding a line that best "fits" the data.
- *For example*: predicting the price of a house using the number of rooms in that house (y: price, x_1 : number of rooms)

Linear Regression

- In general, the equation for linear regression is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p + \epsilon$$

- Where:
  - y: the dependent variable; the thing we are trying to predict.
  - x_i: the independent variables: the features our model uses to model y.
  - β_i: the coefficients (aka "weights") of our regression model. These are the foundations of our model. They are what our model "learns" during optimization.
  - ϵ: the irreducible error in our model. A term that collects together all the unmodeled parts of our data.

Linear Regression

- Fitting a linear regression model is all about finding the set of coefficients that best model y as a function of our features
- Once we've estimated these coefficients, β_i', we predict future values, y', as:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_p x_p$$

Linear Regression

- How It Works

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Let's fit a model to predict housing price
($) in San Diego, USA using the size of the
house (in square-footage):
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$$\text{house-price} = \hat{\beta}_1 * sqft + \hat{\beta}_0$$

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We'll start with a very simple model,
predicting the price of each house to be
just the average house price in our
dataset, ~$290,000, ignoring the different
sizes of each house:
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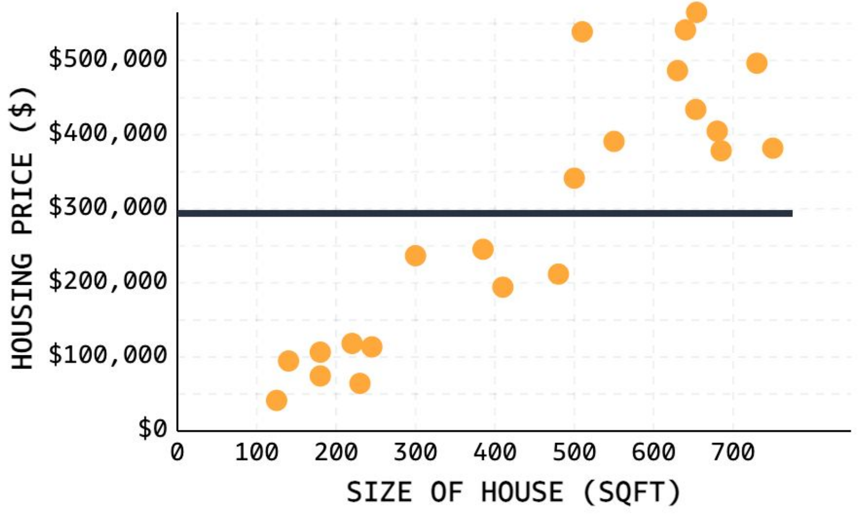
$$\text{house-price} = 0 * sqft + 290000$$

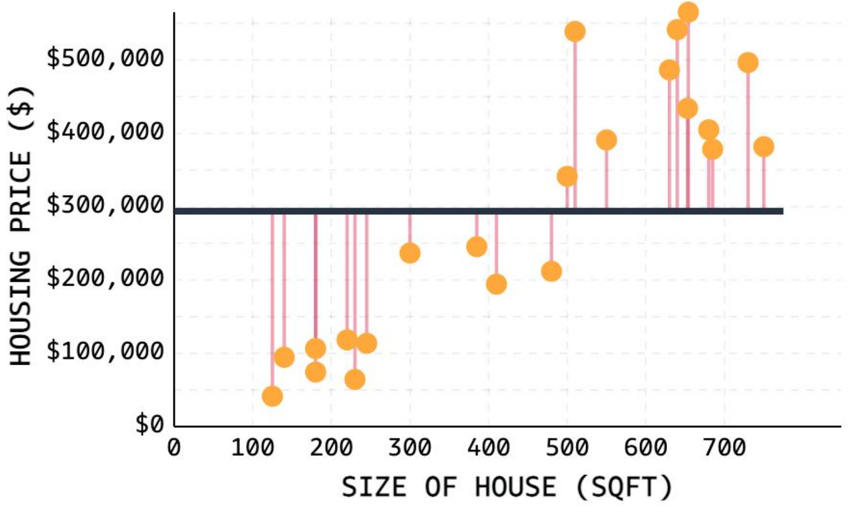


Linear Regression

- How It Works

Of course we know this model is bad - the model doesn't fit the data well at all. But how can do quantify exactly *how* bad?

To evaluate our model's performance quantitatively, we plot the error of each observation directly. These errors, or **residuals**, measure the distance between each observation and the predicted value for that observation. We'll make use of these residuals later when we talk about evaluating regression models, but we can clearly see that our model has a lot of error.

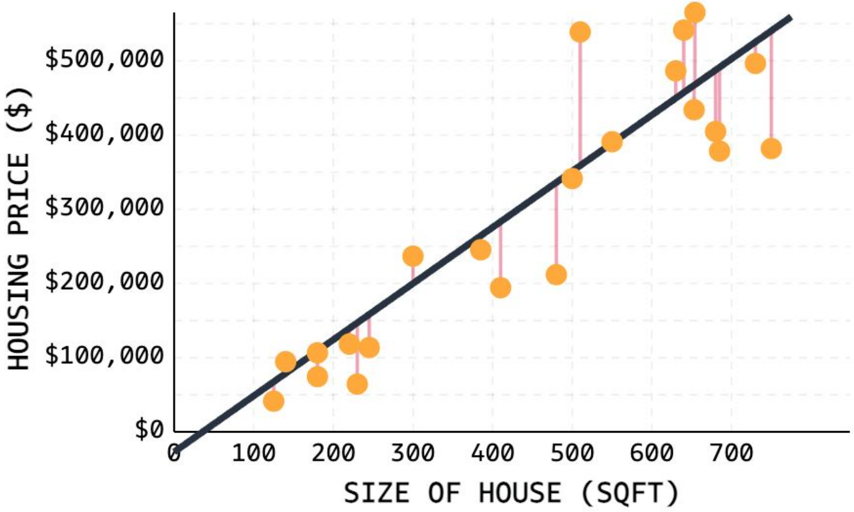


Linear Regression

- How It Works

The goal of linear regression is reducing this error such that we find a line/surface that 'best' fits our data. For our simple regression problem, that involves estimating the y-intercept and slope of our model, $\hat{\beta}_0$ and $\hat{\beta}_1$.

For our specific problem, the best fit line is shown. There's still error, sure, but the general pattern is captured well. As a result, we can be reasonably confident that if we plug in new values of square-footage, our predicted values of price would be reasonably accurate.



Linear Regression

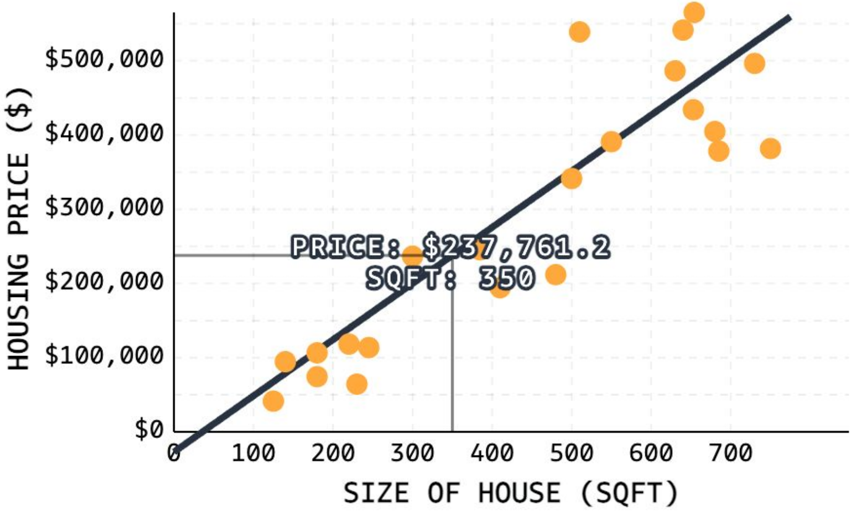
Once we've fit our model, predicting future values is super easy! We just plug in any x_i values into our equation!

For our simple model, that means plugging in a value for $sqft$ into our model:

$sqft$ Value: 350

$$\hat{y} = 756.9 * 350 - 27153.8$$
$$\hat{y} = 237761$$

Thus, our model predicts a house that is 350 square-feet will cost $237,761.



Linear Regression

- Model Evaluation
  - In machine learning, we call such performance-measuring functions loss functions
  - Several popular loss functions exist for regression problems

Linear Regression

- Mean-Squared Error (MSE)
  - MSE quantifies how close a predicted value is to the true value
  - MSE works by squaring the distance between each data point and the regression line, summing the squared values, and then dividing by the number of data points

$$MSE = \frac{1}{n}\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Linear Regression

- R-Squared
    - Regression models may also be evaluated with the so-called goodness of fit measures, which summarize how well a model fits a set of data
    - r-squared measures the percentage of variance explained normalized against the baseline variance of our model

Linear Regression

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- The highest possible value for r-squared is 1, representing a model that captures 100% of the variance.
- A negative r-squared means that our model is doing worse (capturing less variance) than a flat line through mean of our data would.

Linear Regression

- Linear Regression Formula (Calculate by hand)
  - The formula used for linear regressions is, y = a + bx

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

Linear Regression

- Example: find the linear regression equation for the given data

x	y
3	8
9	6
5	4
3	2

x	y	x^2	xy
3	8	9	24
9	6	81	54
5	4	25	20
3	2	9	6
$\Sigma x = 20$	$\Sigma y = 20$	$\Sigma x^2 = 124$	$\Sigma xy = 104$

Linear Regression

- Example: (Cont.)

$$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$a = \{20\ (124) - 20\ (104)\} / \{4\ (124) - 400\}$
$a = 400/96 = 4.17$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$b = \{4\ (104) - 20\ (20)\} / \{4\ (124) - 400\}$
$b = 16/96 = 0.166$

- So, the linear regression equation is, y = 4.17 + 0.166x

Linear Regression

- Example: try to find the regression

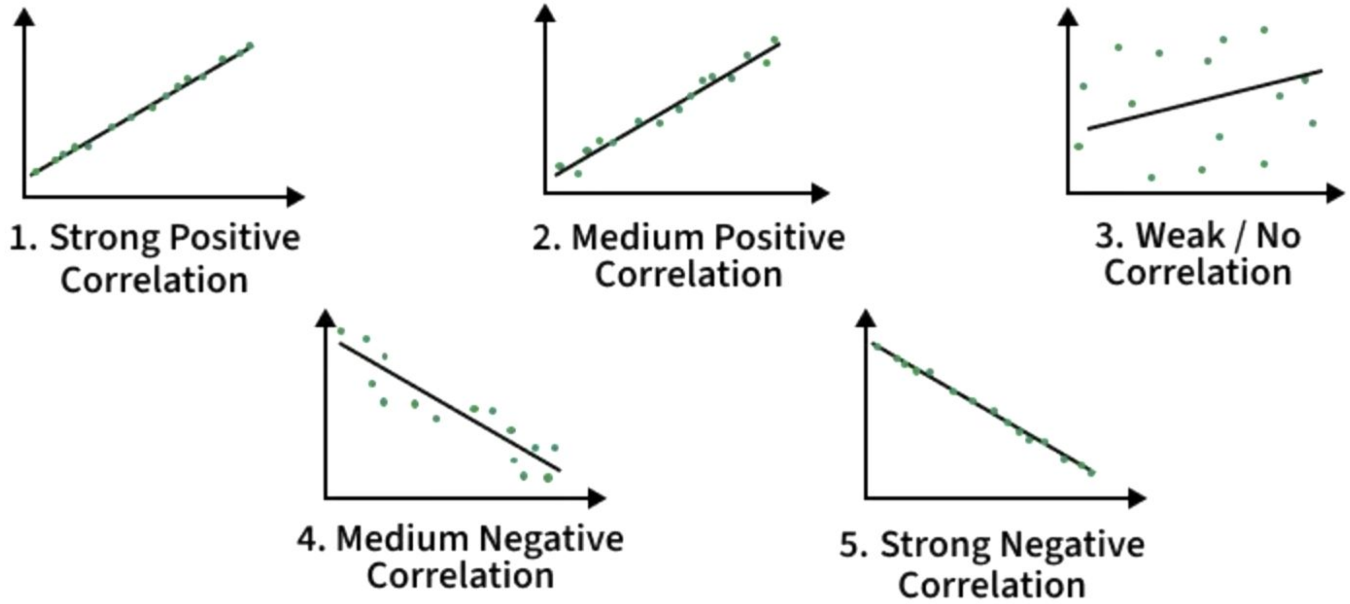
x	y
4	6
7	5
3	8
1	3

Linear Regression

- Pearson Correlation Coefficient
    - It used for measuring the strength and direction of a linear relationship between two variables
    - Value of PCC(denoted by r) ranges from -1 to 1
        - 1 show where both variables increase or decrease together at a constant rate.
        - -1 shows where one variable increases as the other decreases proportionally.
        - 0 shows no linear relationship

Linear Regression

- Pearson Correlation Coefficient



Linear Regression

- Linear Regression Concepts
  - Best Fit: The straight line in a plot that minimizes the divergence between related dispersed data points
  - Coefficient: Also known as a parameter, is the factor that is multiplied by a variable. A coefficient in linear regression represents changes in a Response Variable
  - Coefficient of Determination: It is the correlation coefficient. In a regression, this term is used to define the precision or degree of fit

Linear Regression

- Linear Regression Concepts (Cont.)
  - Correlation: the measurable intensity and degree of association between two variables, often known as the 'degree of correlation.' The values range from -1.0 to 1.0
  - Dependent Feature: A variable represented as y in the slope equation y=ax+b. Also referred to as an Output or a Response
  - Estimated Regression Line: the straight line that best fits a set of randomly distributed data points

Linear Regression

- Linear Regression Concepts (Cont.)
  - Independent Feature: a variable represented by the letter x in the slope equation y=ax+b. Also referred to as an Input or a predictor
  - Intercept: It is the point at where the slope intersects the Y-axis, indicated by the letter b in the slope equation y=ax+b
  - Least Squares: a method for calculating the best fit to data by minimizing the sum of the squares of the discrepancies between observed and estimated values

Linear Regression

- Linear Regression Concepts (Cont.)
  - Residual: the vertical distance between a data point and the regression line
  - Regression Model: The optimum formula for approximating a regression
  - Slope: the steepness of a regression line. The linear relationship between two variables may be defined using slope and intercept: $y=ax+b$

Sklearn Linear Regression

- Scikit-learn (Sklearn) is Python's most useful and robust machine learning package.
- It offers a set of fast tools for machine learning and statistical modeling, such as classification, regression, clustering, and dimensionality reduction

Create a Sklearn Linear Regression Model

- Step 1: Importing All the Required Libraries
- Step 2: Reading the Dataset
- Step 3: Exploring the Data Scatter
- Step 4: Data Cleaning
- Step 5: Training Our Model
- Step 6: Exploring Our Results

Create a Sklearn Linear Regression Model

- Continue in google colab …

End

- Q&A
- Thank you :)